Wraysbury Materials Laboratory

Forensic Engineering Examination Report

Independent examination of guard interlock — machine PT-DB-07

Report reference	WML-26-0418
Prepared for	Whitfield Carter LLP, for Pennington Industries Limited
Examined by	Dr Faisal Iqbal, Principal Metallurgist
Date of examination	12 February 2026
Date of report	18 March 2026
Item examined	Tongue-actuated interlock safety switch and actuator, machine PT-DB-07

1. Instructions and scope

I was instructed by Whitfield Carter LLP, on behalf of Pennington Industries Limited, to examine the interlock safety switch removed from horizontal deburring machine PT-DB-07 following the incident of 14 January 2026, and to determine, so far as possible, the mechanism by which the interlock failed to prevent operation of the cutting spindle while the guard door was open. I am aware of my overriding duty to give independent and objective evidence and this report is prepared on that basis.

2. Background

The guard door of PT-DB-07 was fitted with a tongue-actuated interlock switch of a type in which a shaped actuator (the 'tongue') enters the switch head when the door is closed, holding the safety contacts closed and permitting machine operation; withdrawal of the tongue when the door opens should positively open the contacts and remove the run permit. I received the switch and actuator under continuity of evidence and examined them visually, dimensionally, and by low-power microscopy, together with bench function testing.

3. Findings

3.1 The actuator tongue exhibited significant wear and plastic deformation at its leading faces, with rounding of the engagement profile consistent with prolonged operation, misalignment of the door, and repeated forced entry. Material loss reduced the effective length of the tongue.

3.2 Within the switch head, the cam mechanism that should positively drive the contacts open on withdrawal of the tongue was found to be worn and contaminated with fine swarf and coolant residue. On bench testing, the safety contacts did not open reliably as the actuator was withdrawn: in a proportion of cycles the contacts remained closed, leaving a run permit present with the actuator fully withdrawn (i.e. door open).

3.3 The switch is of a category that, when worn or contaminated in the manner found, can fail in a manner that does not achieve positive opening. The combination of tongue wear and internal contamination defeated the interlock such that the spindle could be commanded to run with the guard open.

4. Opinion on mechanism of failure

In my opinion, the interlock failed because the tongue-actuated switch did not achieve positive opening of its safety contacts on withdrawal of the actuator, as a result of cumulative mechanical wear of the actuator and switch cam together with ingress of swarf and coolant. The failure mode is one that periodic function testing is specifically designed to detect: a simple test (opening the guard and confirming that the spindle cannot be started or continues to be inhibited) would, in my view, have revealed the defect. Replacement of the worn switch, or routine function testing at appropriate intervals, would on the balance of probabilities have prevented the dangerous condition from persisting undetected.

5. Recommendation

Tongue-actuated interlocks in this dusty, swarf-laden environment are vulnerable to the wear and contamination found. I recommend their replacement on safety-critical guards with coded, non-contact (e.g. RFID-coded magnetic) safety switches, which are markedly more tolerant of contamination and resistant to defeat, supported by a documented periodic function-test regime.

Dr Faisal Iqbal CEng — Principal Metallurgist, Wraysbury Materials Laboratory

I confirm that this report is given in accordance with my duty to provide independent expert evidence and that the opinions expressed are my own.